



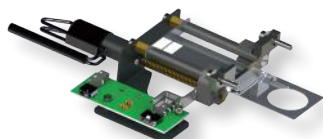
Continuous, Automatic Suspended Particulate Monitor MP101M-LCD



Complete measurement system with
RSP sampling tube and sampling
head



TCP/IP remote ESA Connect™
interface with animated diagram and
intuitive navigation by pictograms



Automatic and programmable
Span Calibration Module

The MP101M analyzer uses the standard ISO 10473 by beta gauge measurement method for the continuous measurement of fine dust in ambient air.

The instrument is certified equivalent to the reference method in accordance with the EN12341 European standards for PM10 and PM2.5 respectively EN14907 and US EPA for the continuous monitoring of PM10 or PM2.5.

SPECIFIC FEATURES:

- High accuracy, not influenced by the physicochemical nature, color or shape of particles
- Ultra-low activity C¹⁴ sealed flat source
- True volumetric air flow control with 3 atmospheric pressure and temperature sensors
- Sampling flow-rate continuously regulated to the atmospheric temperature and pressure: reduces evaporation artefacts of volatile compounds (mandatory for PM2.5 according to EU regulations)
- Built-in reference gauge for calibration: no need for factory re calibration
- Regulated Sampling Tube (RST) compliant with CEN PM10 and US-EPA standard: the sample is not affected by seasonal or geographical factors (losses of semi-volatile particles by evaporation)
- Fibreglass tape **with 3 years of autonomy** of continuous daily sampling
- Rugged instrument, not sensitive to vibration, humidity, temperature...
- Data recording through RS232, Ethernet, USB port (included in standard)
- Extremely compact: easy to install as bench top or rack mounting in mobile or stationary laboratories
- Optionally: external CPM module for real time particulate counting and mass concentration measurement
- Interchangeable sampling heads PM10, PM2.5, PM1 and TSP
- **New:** On board web server compatible with any internet browser; ESA Connect™ user interface with on-line help for display, configuration, maintenance, diagnostics or software updating, remotely, from any PC, tablet or smartphone.
- Special module for automatic calibration (option)

CERTIFICATIONS / STANDARDS:

- > Standard compliance: **ISO 10473 : 2000**
- > **for PM10:**
 - **US-EPA** (EQPM-0404-151)
 - **EN 12341** (I-CNR 087/2004, F-LCSQA)
- > **for PM2.5:**
 - **EN 14907** (F-LCSQA)
 - **US-EPA** (EQPM-1013-211)
 - **J-MOE PM 2.5** Type approved



Suspended Particulate Monitor MP101M-LCD

SPECIFICATIONS:

- Measurement ranges: user-programmable, up to **10 000 $\mu\text{g}/\text{m}^3$**
- Lower detectable limit: $0.5 \mu\text{g}/\text{m}^3$ (24h average)
- Measurement cycles: 1/2, 1, 2, 3, 6, 12, 24 h, user-selectable (up to 96 hours)
- Measuring period: 10 min, 1/4, 1/2, 1, 2, ..., 48h (user-selectable)
- Beta Source: sealed Carbon 14 ($1.6\text{MBq} \pm 15\%$)
- Detector: high performance Geiger-Müller counter
- Sampling flow rate: $1 \text{ m}^3/\text{h}$
- Standard filter: fibreglass tape (width 35 mm, length 30 m); autonomy for 1,200 samples (>3 years of daily measurements)
- Power supply: 230V/50Hz (115V/60 Hz)
- Housing: 19" rack / 6U
- Dimensions: 483 x 324 x 266 mm (W x D x H)
- Weight: 15 kg (without pump)
- Operating temperature: +10 to +40 °C
- Serial link: 1 RS 232/RS422
- Ethernet (RJ45) and USB ports
- On-board webserver with remote ESA Connect™ interface included

OPTIONS AND ACCESSORIES:

- CPM module** for optical real-time measurement (concentration, counting, classification by size range) of particles (see specific brochure)
- US EPA and EU-CEN compliant sampling inlets
- Temperature-regulated sampling tube (RST):** 1 m, 1.5 m, 2 m, 2.75 m, compliant with **CEN PM10 Directive**
- Max 2 ESTEL electronic boards with: 4 independent analog inputs / outputs, 4 remote control inputs, 6 dry contacts outputs
- External pump assembly (diaphragm - 9,5 kg or rotary vane - 7 kg)
- All-weather cabinet for outdoor use, with or without air conditioning
- Easy to install span calibration module for automatic and programmable calibrations
- Field connection kit for leak and zero test (on RST tube)
- Laboratory connection kit for leak and zero test (on MP101)
- Bead flow meter for leak test
- HEPA filter for zero test



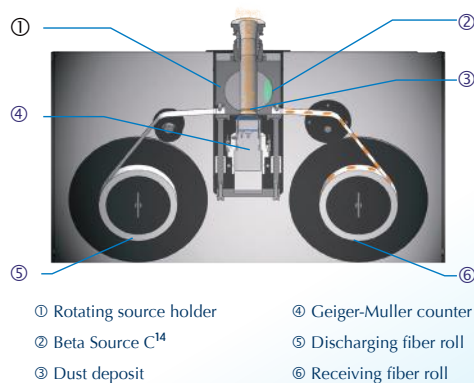
Sliding drawer on the rear panel for easy access and maintenance

MEASUREMENT PRINCIPLE:

The MP101M is a measuring device of suspended particles in the ambient air. It allows regulatory oversight of PM10 and PM2.5, with an alarm in case of threshold exceeding. The analyzer is based on the beta ray attenuation measurement technique; it determines the particles concentration by measuring the amount of radiation that a sample, collected on a fiber tape, absorbs when exposed to a radioactive source. Low energy beta rays are absorbed by collision with electrons, whose number is proportional to density. Absorption is thus a function of the mass of the irradiated material, independently of its physicochemical nature.

Combined with the optional CPM (Continuous Particulate Measurement) module, the monitor allows a precise and real-time monitoring of PM10, PM2.5 and PM1 simultaneously. The CPM module can be plugged additionally to all existing MP101M, between the RST sampling line and the MP101M monitor.

MP101M - operating principle



Distributed by:



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